Jorge Manuel Torre

Ashburn, VA • jorgetorre06@gmail.com • 571-888-4832 • linkedin.com/in/jmt1006 • github.com/jmtorr3 • jmtorr3.github.io

Education

Virginia Tech — Blacksburg, VA
B.S. Computer Science, Overall GPA: 3.32/4.0, Major GPA: 3.50/4.0 — May 2026
Relevant Coursework: Data Structures & Algorithms, Computer Systems, Computer Architecture, Software Reverse Engineering, Linux Kernel Programming, Security of AI.

Projects & Experience

Garage64: Custom Entity Model Studio • GitHub — August 2025 – Present
React, Three.js, Django, SQLite, Docker
- Built a self-hostable full-stack web suite for authoring custom Minecraft car models (OptiFine CEM) and texture packs, packaged with Docker Compose for reproducible local setup.
- Engineered a React and Three.js frontend to provide a real-time 3D workspace, allowing users to dynamically compose vehicles by swapping parts and applying custom paint textures.
- Backend powered by a Django REST API for entity data management, with a one-click export pipeline producing production-ready .jem and .jpm asset files.

personal-blog: Full-Stack Blogging Platform • GitHub — February 2026 – April 2026
React, Vite, Django REST Framework, PostgreSQL, Docker, Nginx
- Designed and built a fork-friendly blogging platform with a Django REST API backend and a React + Vite frontend, containerized end-to-end with Docker Compose and an Nginx reverse proxy.
- Implemented JWT-authenticated admin (SimpleJWT) with a draft/publish workflow, cover image upload, and live-preview markdown editing using react-markdown and syntax highlighting.
- Designed a public read-only API for syndication into a separate portfolio site, structured so other developers can fork the repo and customize.

CHUD: Catastrophic Harmful Update Detection • GitHub — January 2026 – April 2026
Python, PyTorch, LoRA, Llama-2-7b, Llama-Guard-3-8b, Unsloth — *Team of 4 (CS Capstone)*
- Investigated weaknesses in the LoX defense (Perin et al., COLM 2025) via an iterative LoRA fine-tuning attack on Llama-2-7b that interleaves benign GSM8k and adversarial BeaverTails samples to drive catastrophic forgetting in safety-critical subspaces.
- Built an Attack Success Rate (ASR) evaluation pipeline using Llama-Guard-3-8b as a judge model, calibrated against AdvBench (99% accuracy) and BeaverTails; ran 30+ ASR experiments across attack mixes and learning-rate sweeps.
- Identified a robustness gap in LoX: direct harmful fine-tuning achieves 91–96% ASR on Chat-aligned Llama-2, LoX reduces this to 37–63% under direct attack, but iterative mixed-data attack restores 89% ASR — flattening the safety landscape without eliminating exploitability.
- Provisioned dual VT GPU cluster environments (x86-64 and aarch64) for the team's training pipeline; owned QLoRA fine-tuning evaluation and integrated results into the final ASR analysis.

Celeste-RL: Deep Reinforcement Learning for a Precision Platformer • GitHub • Website — January 2026 – April 2026
Python, PyTorch, NumPy, matplotlib, CUDA
- Conducted a controlled comparison of five RL methods (DQN, Behavioral Cloning, Hybrid, Curriculum, Dueling DQN with curiosity) trained to play Celeste Classic on the Pyleste emulator.
- Shipped a reproducible pipeline with batched evaluation, automated comparison plots, gameplay GIF generation, and a GitHub-Pages-deployed results writeup covering methodology and findings.

Experimental Home Lab — January 2021 – Present
Proxmox, Debian, Ubuntu, NixOS, Docker
- Architected a Proxmox-based home lab with OPNsense firewall, hardened SSH jumpbox access, and reproducible NixOS / Docker environments, enabling rapid prototyping of self-hosted services and low-level systems experimentation.

Leadership & Community

Team Lead, Database Management Systems (CS 4604) — May 2025 – August 2025
- Directed a 3-person team in collaborative schema design and SQL development, facilitating design discussions and integrating individual contributions into a unified submission.

Team Lead, Intermediate Software Design (CS 3704) — January 2025 – April 2025
- Led a 5-person team through the full software lifecycle including requirements gathering, sprint planning, and code reviews; coordinated deliverables across team members with differing schedules and skill levels.

Garage32 (Community Project) — March 2020 – December 2021
- Led development and distribution of a graphical asset pack for the Minecraft community, reaching **600+ downloads**; promotional content earned **30K+ likes** on a single TikTok post.

Skills

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL, Bash
Web & Frameworks: React, Three.js, Vite, Django, Django REST Framework, Node.js, REST APIs
Infrastructure: PostgreSQL, Docker, Nginx, Linux, Git, CI/CD, NixOS, Proxmox